

Leveraging Short-Term, Map-Specific Player Statistics for Counter-Strike 2 Match Prediction

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Abstract

This project explores the prediction of Counter-Strike 2 (CS2) match outcomes using machine learning techniques, with a focus on short-term (90-day), map-specific player performance metrics. We introduce a novel feature representation strategy and evaluate the performance of TabPFN - a Transformer-based meta-learning model - against tuned classical models including logistic regression, XGBoost, and LightGBM. Our findings suggest that short-term player statistics enhance predictive performance, achieving test accuracy and AUC scores that exceed previously published models. These results have implications for algorithmic betting and the broader field of esports analytics.

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1 Introduction

The primary objective of this project is to develop a predictive model for Counter-Strike 2 match outcomes that outperforms existing public models. The central research question is: Can a novel data representation strategy, combined with the state-of-the-art TabPFN model, yield significantly higher predictive accuracy on HLTV-derived Counter-Strike 2 data than existing models?

Our approach centres on restructuring input data to emphasize recent player performance on a per-map basis, rather than relying on long-term aggregate statistics. This refinement is motivated by the hypothesis that short-term, map-specific performance trends are more predictive of upcoming match outcomes, particularly in the context of Counter-Strike 2, where team dynamics and player form can vary substantially across space and time.

Although prior work has explored Counter-Strike: Global-Offensive and Counter-Strike 2 match prediction using conventional machine learning models, to the best of our knowledge, no study has yet evaluated the performance of TabPFN - a recently released Transformer-based probabilistic meta-learning model - on esports data. This project introduces novelty in both model selection and evaluation: we benchmark TabPFN against a tuned ensemble of classical models, including logistic regression, XGBoost, and LightGBM. This modelling strategy, combined with our domain-specific feature engineering - centered on short-term, map-aware player metrics - represents a methodological contribution to the field of Counter-Strike 2 esports analytics.

The primary impact of this work lies in its potential to improve predictive accuracy for Counter-Strike 2 match outcomes using publicly available data. Enhanced model performance could support the development of algorithmic betting strategies, where even marginal gains in prediction accuracy may translate to financial returns. More broadly, the project contributes to the growing field of data-driven esports analysis by applying recent advances in meta-learning to a novel domain and by demonstrating the value of performance-aware feature construction in match forecasting.

The problem we address - accurate prediction of CS2 match outcomes - is relevant not only to bettors, but also to analysts, teams, and fans who rely on performance forecasting. Improved prediction models have practical, financial, and strategic implications, underscoring the value of this investigation.

2 Background

Counter-Strike 2 (CS2) is a competitive round-based first-person shooter that serves as the successor to Counter-Strike: Global Offensive (CS:GO). In professional esports settings, CS2 matches are played in structured formats such as best-of-one (Bo1), best-of-three (Bo3), or best-of-five (Bo5) series, where teams compete to be the first to win 13 rounds and thus earning them an individual map win. A central feature of competitive CS2 is its map pool - a curated set of playable maps each having a unique geometric layout allowing for different play-styles of players to shine through. Thus, team's or player's effectiveness is not solely a function of their overall skill, but also of their compatibility with the specific tactical demands of a given map. This gives rise to the concept of map-specific performance which takes into account not the aggregated statistics of a player across all recent maps but rather a specific one. Typical metrics used to characterize player performance include: kills, deaths, kill/death ratio, headshot percentage, damage per round, and their overall rating (Rating 2.0) which is calculated by HLTV using above metrics and many more.

The required data needed for this project is largely accessible through HLTV, a news website and forum which covers professional CS2 esports news, tournaments, and statistics such as individual player performance metrics and team performance metrics. Historical betting odds,

which would serve as a valuable benchmark for evaluating practical effectiveness, are not fully available. While betting odds for upcoming matches can be accessed online - in fact HLTV lists them on their website - the lack of automated web-scraping experience presents a practical limitation that makes full evaluation through simulated betting (paper trading) difficult.

3 Literature Review

Previous studies have explored the use of machine learning to predict outcomes of professional Counter-Strike matches, focusing primarily on CS:GO - the predecessor to the more recently released CS2 - by leveraging statistical data from platforms such as HLTV. One of the most comprehensive studies is by Tabacof [1], who developed a model using LightGBM trained on over 30,000 professional matches with extensive feature engineering, including Bayesian TrueSkill ratings. His approach achieved high predictive performance with a reported accuracy of 70.1% and AUC of 0.773 on an out-of-time test set, making it one of the most effective pre-match models to date. The study included a brief feature importance analysis although no actual feature selection in the form of permutation importance or adversarial validation was performed.

Similarly, Švec [2] conducted an extensive comparison of models using a dataset of over 40,000 matches. He evaluated several sample representations and models, including linear models, tree ensembles, and deep learning architectures. His best-performing model was based on a player-level Elo rating system, which achieved a test accuracy of 64%, closely followed by a Random Forest model at 63%. Švec’s work also emphasized the importance of using round-based team and player statistics, and he experimented with CNNs and embedding-based neural networks, although their performance did not surpass simpler models.

Johansson & Källström [3] focused on logistic regression and tree-based models using a curated dataset of 3,788 matches from 2016-2020. Their LASSO-regularized logistic regression model achieved an accuracy of 60.5%, with a Cohen’s kappa of 0.21. Their evaluation also included a direct comparison with bookmaker odds during the BLAST Premier World Final 2023, where their random forest model achieved an accuracy of 55.6%, compared to 77.8% accuracy for the implied betting market.

Other studies such as those by Makarov et al. [4] and Xenopoulos et al. [5] have also contributed valuable insights, with models incorporating probabilistic skill ratings and in-game event streams achieving accuracy of 62% and AUCs as high as 0.791. Overall, the literature consistently finds that performance metrics such as Elo, TrueSkill, and well-curated historical features are among the strongest predictors of match outcomes, with most models achieving predictive accuracies between 60% and 65%.

4 Data and Feature Engineering

The dataset used for this project was obtained via Kaggle [6], covering professional Counter-Strike 2 matches played between November 2023 and January 2024. In addition to match identifiers (matchID), team names, and the specific map played, the raw dataset included detailed player-level performance metrics for all ten participants, alongside team-level map win-rates. Crucially, all provided player and team statistics reflect performance specifically on the map being played, aggregated over the preceding 90 days.

Initial data cleaning involved removing rows with missing map information and excluding columns corresponding to the obsolete Rating 1.0 system (a legacy feature from CS:GO, superseded by Rating 2.0 in CS2). This preprocessing resulted in a dataset containing 1010 unique matches, described by 147 initial features and a binary target column indicating the match winner. Of the 147 features, 140 represented player-specific statistics (14 distinct metrics per player), and two represented the map-specific win rates for each team. The 14 player-level metrics captured various aspects of performance, including: total kills, headshot percentage, total

deaths, kill/death ratio, damage per round, grenade damage per round, maps played, rounds played, kills per round, assists per round, deaths per round, saves by teammate per round, saved teammates per round, and the overall HLTV Rating (Rating 2.0). To create team-level representations capturing both aggregate performance and intra-team consistency, the 14 types of individual player statistics were aggregated within each five-player team. Specifically, for each metric type (e.g., kills, kill/death ratio), five summary statistics were computed across the team’s players: the mean, standard deviation, sum, minimum, and maximum values. This aggregation process generated 70 derived statistical features per team (14 metrics with 5 aggregations per each), resulting in 140 team-based features across both competing teams per match. These 140 derived features were then combined with the two original team-specific map win-rate features.

Consequently, the final dataset prepared for modelling consisted of 1010 rows, each described by 142 predictive features. The target variable was encoded as a binary column, where a value of 1 indicated a win for the designated "Team 1", and 0 indicated a loss. This finalized dataset was subsequently partitioned into an 80% training set and a 20% test set to facilitate model training and evaluation.

4.1 Preliminary Results

To obtain an initial assessment of the predictive potential of the engineered features, a simple, untuned logistic regression model was trained. Evaluating this model on the test set it achieved an accuracy of 70.3% and an AUC score of 0.782. A result which is similar to that of the author of the Kaggle dataset which achieved an accuracy of 71% [6]. These preliminary results were notably strong exceeding accuracies reported in prior CS:GO/CS2 prediction studies. While these findings suggest the effectiveness of our feature representation strategy, particularly the focus on short-term, map-specific metrics, caution is warranted regarding direct comparison due to the dataset’s limited size (1010 samples) and specific temporal scope (November 2023 - January 2024). A 95% confidence interval for the accuracy was calculated as [0.6750, 0.7310], indicating again promising results. Despite these considerations and the need for further validation, the encouraging preliminary performance motivated the decision to proceed with training more increasingly complex models, such as XGBoost, LightGBM, and TabPFN, utilizing the developed feature set.

5 Research Plan and Methodology

This study employs a comparative machine learning approach to evaluate predictive models for CS2 match outcomes, utilizing the feature set developed previously. The core methodology involves training and comparing several models, including both a baseline logistic regression and a version optimized via cross-validated hyperparameter tuning, exploring techniques such as regularization. Similarly, gradient boosting machines, specifically XGBoost and LightGBM, will be evaluated by comparing baseline configurations against versions tuned using cross-validation. Furthermore, the state-of-the-art Transformer-based meta-learning model, TabPFN, will be applied to assess its performance on this specific tabular esports dataset. The research process commences with the data preparation steps of cleaning the dataset and implementing the feature engineering strategy detailed in Section 4. Following this, the selected models are trained on the training partition of the data, incorporating cross-validation on this partition to determine optimal hyperparameters for the tunable models (Logistic Regression, XGBoost, LightGBM). Subsequently, all trained models are evaluated on the held-out test set using standard classification metrics, namely Accuracy, AUC, Log-Loss, and Average Precision. Feature importance analysis will then be conducted using methods like L1 regularization coefficients or permutation importance, these insights may inform model refinement, potentially involving feature selection and retraining. The final stage involves a comparative analysis, benchmarking the performance

of the developed models against each other. Contingent on successful data acquisition via web-scraping, an additional evaluation against historical betting odds will be performed as an external reference point.

6 Results and Analysis

6.1 Logistic Regression

A baseline logistic regression model was compared against a version optimized using cross-validated hyperparameter tuning. The tuning process identified L1 regularization (Lasso) as the most effective strategy. Lasso regularization performs inherent feature selection by shrinking the coefficients of less informative features towards zero. Consequently, examining the non-zero coefficients in the tuned model provides insight into feature relevance. Key predictors identified through this process included player Rating 2.0, deaths per round, and team-specific map win rates (Figure 1).

As shown in Table 1, the tuned model exhibited superior performance on the test set across all evaluation metrics compared to the baseline. Notably, accuracy increased by 3 percentage points, and improvements were also observed in the Area Under the ROC Curve (AUC) and Average Precision (AP), visualized in Figure 2. Log-loss saw a marginal improvement, decreasing slightly.

Furthermore, analysis of the confusion matrices (Figure 3) revealed a significant reduction in the number of false positives (incorrectly predicting a win for "Team 1") achieved by the tuned model. This reduction is particularly beneficial in contexts like algorithmic betting, where the cost of a false positive (betting on a predicted win that results in a loss) can outweigh the missed opportunity of a false negative (not betting on an actual win).

Model	Accuracy	AUC	Log-Loss	AP
Basic Logistic Regressor	0.703	0.782	0.559	0.826
Tuned Logistic Regressor	0.733	0.801	0.548	0.847

Table 1: Performance comparison of a basic and tuned logistic regression model on the test set, evaluated using Accuracy, AUC, Log-Loss, and AP.

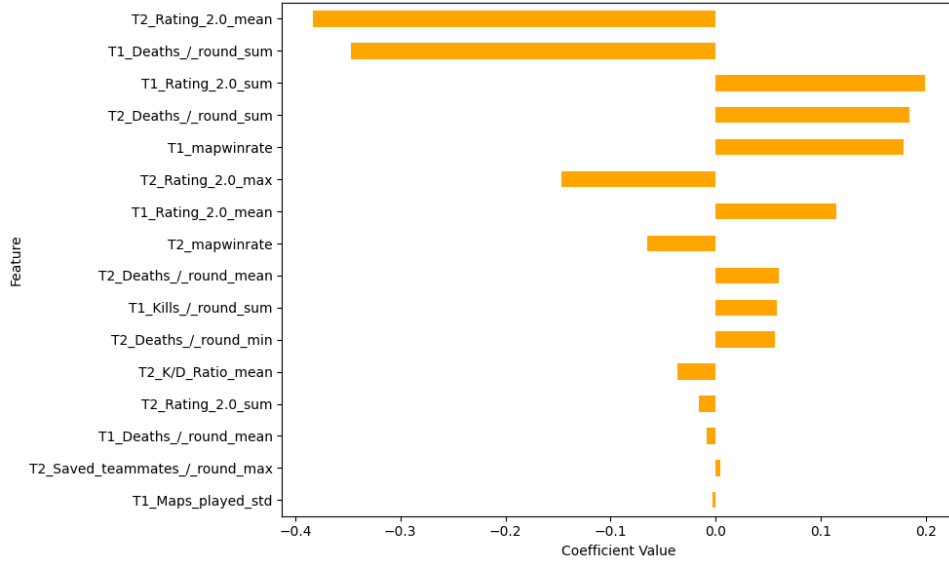


Figure 1: Non-zero feature coefficients from the tuned L1-regularized (Lasso) logistic regression model. Features with positive coefficients increase the predicted probability of Team 1 winning, while negative coefficients decrease it.

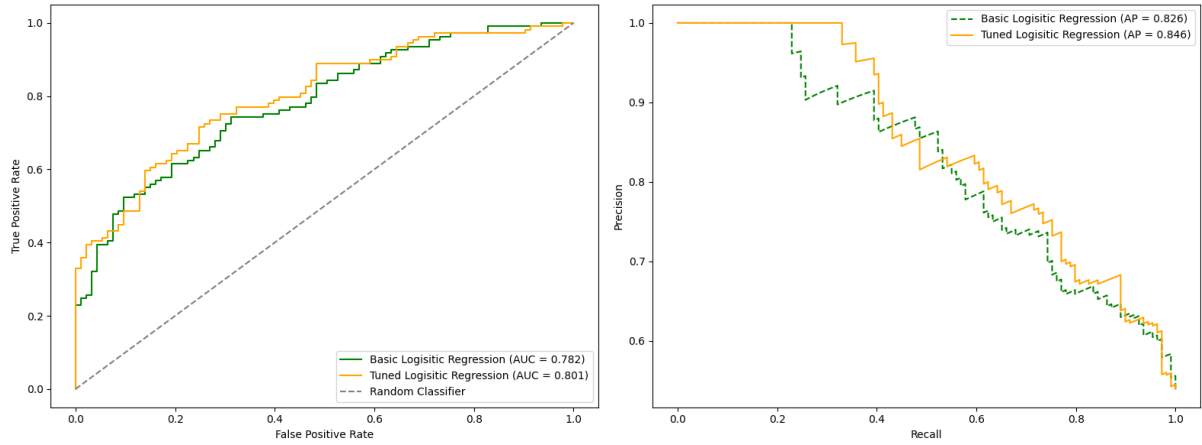


Figure 2: Performance curves comparing basic and tuned (Lasso) logistic regression models on the test set. Left: Receiver Operating Characteristic (ROC) curves showing True Positive Rate vs. False Positive Rate. Right: Precision-Recall (PR) curves. Area Under Curve (AUC) and Average Precision (AP) scores are indicated in the legends.

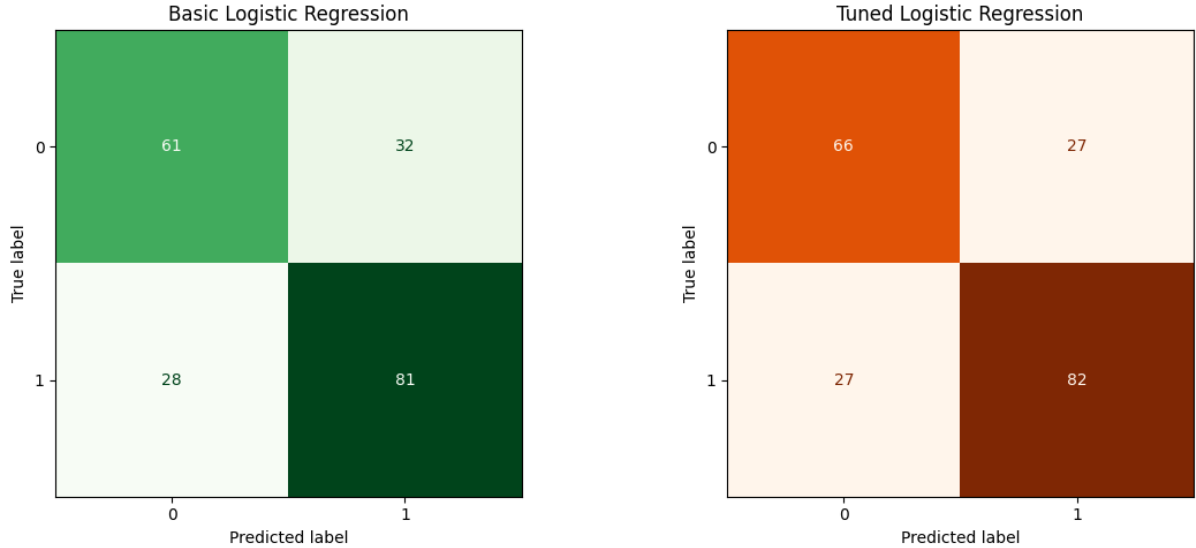


Figure 3: Comparison of confusion matrices for the basic (left) and tuned (Lasso, right) logistic regression models on the test set. Values indicate counts for True Negatives (top-left), False Positives (top-right), False Negatives (bottom-left), and True Positives (bottom-right).

6.2 LightGBM

The performance of three LightGBM model variations was evaluated: a baseline configuration, a version tuned using cross-validated hyperparameter optimization, and a final version trained on a pruned feature set identified via permutation importance.

Compared to the baseline model, hyperparameter tuning yielded slight improvements in Accuracy, AUC, and AP on the test set, although it resulted in a higher Log-Loss (Table 2 and Figure 4). To reduce model complexity, permutation feature importance was calculated using the tuned model. Features contributing below a predefined importance threshold were subsequently removed.

A new LightGBM model was then trained using only this reduced feature set. This 'pruned' model demonstrated markedly superior performance compared to both the basic and tuned versions across all metrics, achieving the highest Accuracy, AUC, and AP, along with the lowest Log-Loss (Table 2). The confusion matrices further illustrate this enhanced predictive capability, showing a clear increase in correctly classified instances and a reduction in false positives for the pruned model (Figure 5).

Analysis of the feature importances derived from the permutation process (Figure 6) indicated that the kill/death ratio was a highly influential predictor, despite exhibiting considerable variability across permutations reflected in its standard deviation. Other features demonstrating significant predictive value included the teams' map-specific win rates and deaths per round.

Model	Accuracy	AUC	Log-Loss	AP
Basic LightGBM	0.728	0.814	0.657	0.844
Tuned LightGBM	0.733	0.815	0.746	0.852
Pruned LightGBM	0.797	0.862	0.569	0.889

Table 2: Performance comparison of a basic, tuned, and pruned LightGBM model on the test set, evaluated using Accuracy, AUC, Log-Loss, and AP.

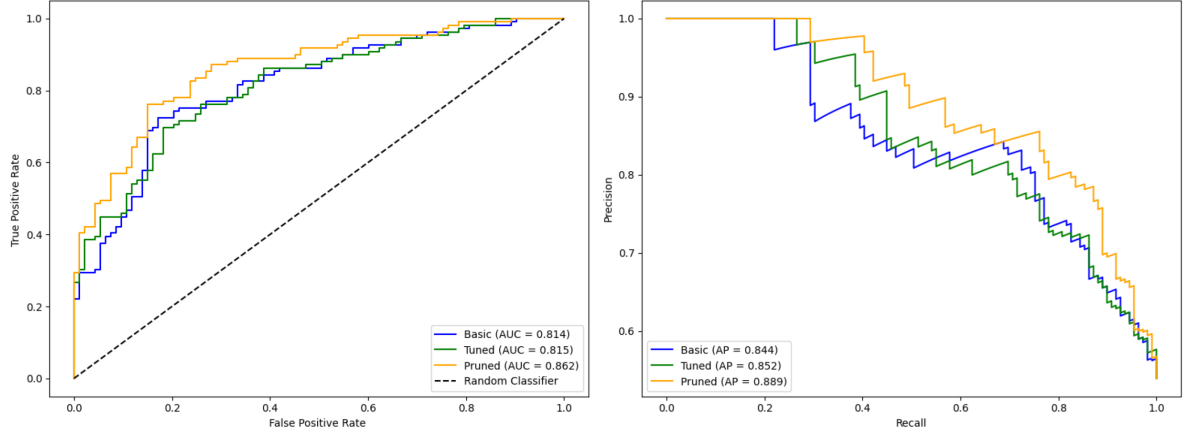


Figure 4: Performance curves comparing basic, tuned, and pruned LightGBM models on the test set. Left: ROC curves. Right: PR curves. AUC and AP scores are indicated in the legends for each model variation.

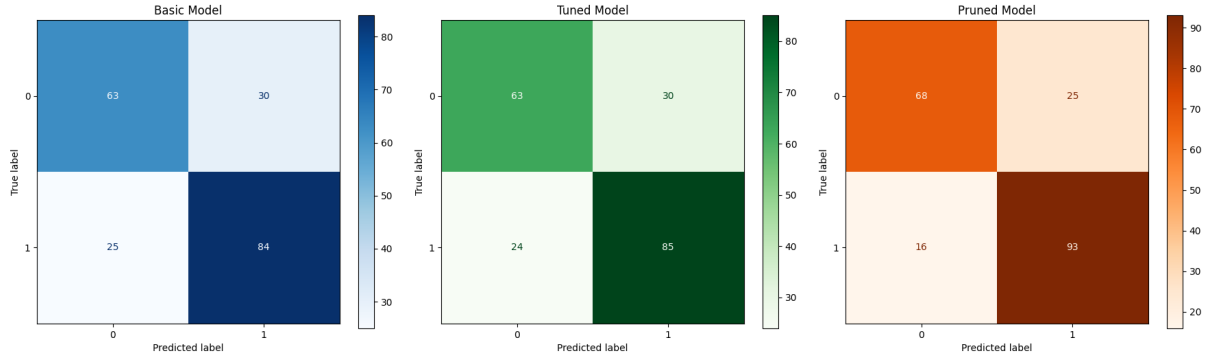


Figure 5: Comparison of confusion matrices for the basic (left), tuned (center), and pruned (right) LightGBM models on the test set. Values indicate counts for True Negatives (top-left), False Positives (top-right), False Negatives (bottom-left), and True Positives (bottom-right).

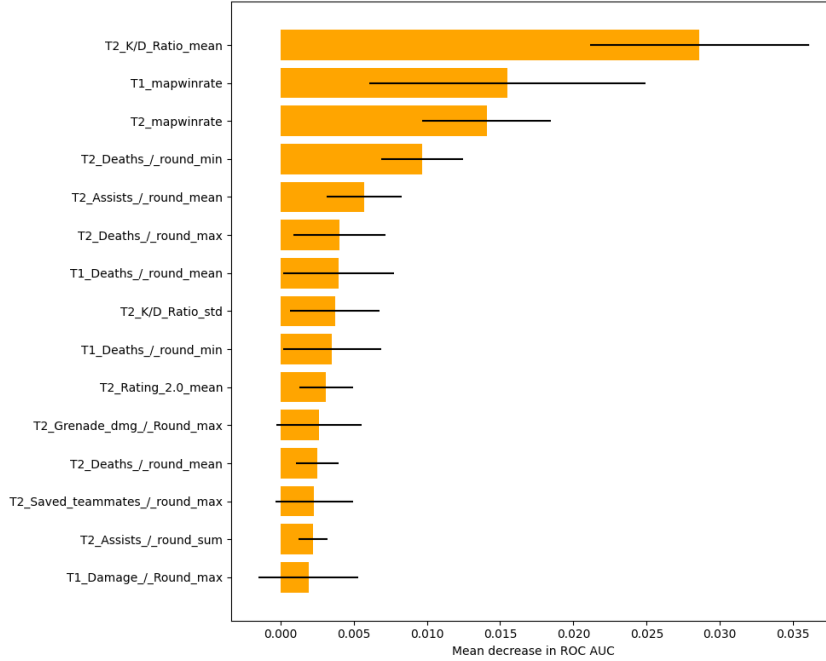


Figure 6: Top 15 features ranked by permutation importance for the tuned LightGBM model. Importance is measured as the mean decrease in test set ROC AUC when the feature’s values are randomly permuted. Error bars indicate the standard deviation across permutations.

6.3 XGBoost

Evaluation of the XGBoost models followed a similar three-stage process: comparing a baseline configuration, a version optimized via cross-validated hyperparameter tuning, and a final model trained on a pruned feature set.

As detailed in Table 3, hyperparameter tuning improved the model’s performance over the baseline on AUC, Log-Loss, and AP when evaluated on the test set (Figure 7). However, this came at the cost of a slight decrease in accuracy.

To explore potential improvements through feature selection, permutation importance was computed using the tuned model. Features falling below a set importance threshold were subsequently removed. An XGBoost model retrained using only this reduced feature set achieved the highest performance across all evaluated metrics, outperforming both the basic and tuned configurations with a test accuracy of 77.7% and an AUC of 0.834.

The feature importance analysis itself highlighted team-specific map win-rates, deaths per round, and kill/death ratio as being among the most influential predictors for the XGBoost model (Figure 8). Correspondingly, the confusion matrix for the pruned model demonstrated a pronounced increase in correct classifications compared to the non-pruned versions, accompanied by a modest reduction in the number of false positives (Figure 9).

Model	Accuracy	AUC	Log-Loss	AP
Basic XGBoost	0.733	0.799	0.558	0.830
Tuned XGBoost	0.719	0.815	0.536	0.850
Pruned XGBoost	0.777	0.834	0.513	0.871

Table 3: Performance comparison of a basic, tuned, and pruned XGBoost model on the test set, evaluated using Accuracy, AUC, Log-Loss, and AP.

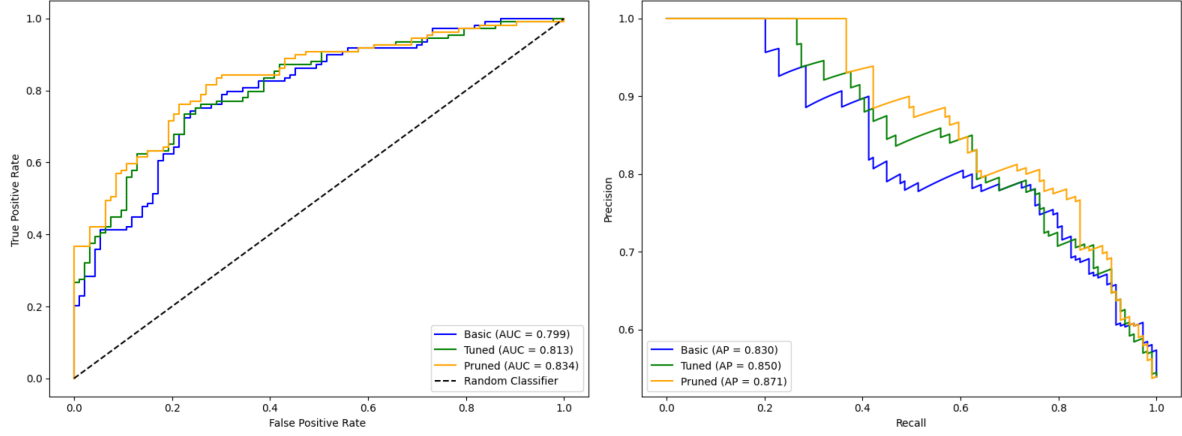


Figure 7: Performance curves comparing basic, tuned, and pruned XGBoost models on the test set. Left: ROC curves. Right: PR curves. AUC and AP scores are indicated in the legends for each model variation.

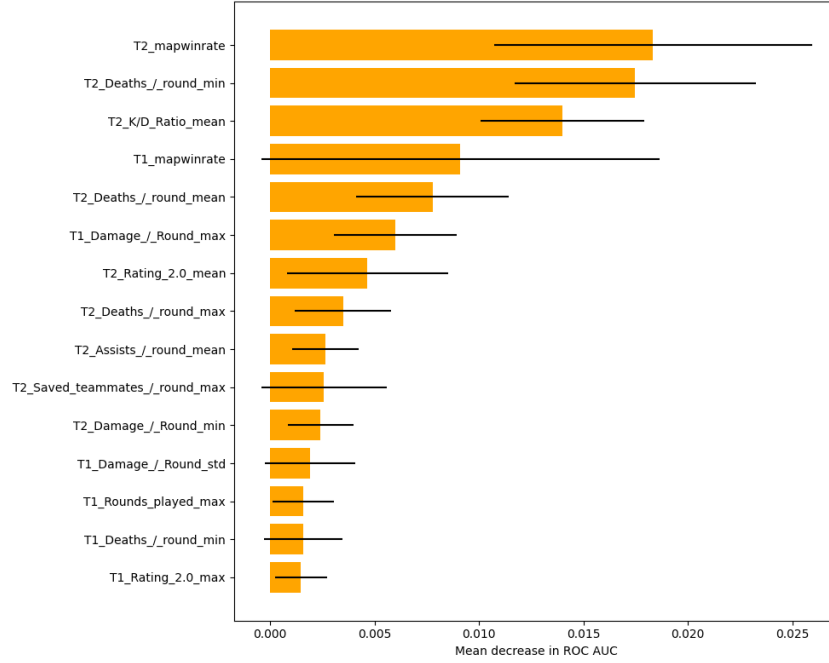


Figure 8: Top 15 features ranked by permutation importance for the tuned XGBoost model. Importance is measured as the mean decrease in test set ROC AUC when the feature's values are randomly permuted. Error bars show the standard deviation across permutations.

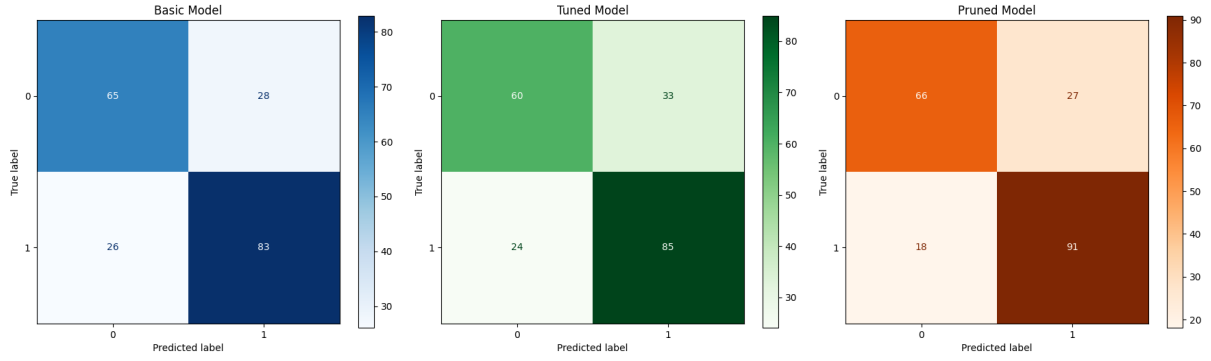


Figure 9: Comparison of confusion matrices for the basic (left), tuned (centre), and pruned (right) XGBoost models on the test set. Values indicate counts for True Negatives (top-left), False Positives (top-right), False Negatives (bottom-left), and True Positives (bottom-right).

6.4 TabPFN

TabPFN operates differently from the tree-based models evaluated, as a pre-trained transformer designed for tabular data, it achieves strong performance without requiring hyperparameter tuning [7]. We evaluated TabPFN under four distinct scenarios: firstly, using the complete set of 142 engineered features, and subsequently, using three pruned feature subsets derived from the feature importance analyses previously conducted for the tuned Logistic Regression, LightGBM, and XGBoost models.

The performance of these TabPFN configurations on the test set is detailed in Table 4. Employing the feature subsets derived from Logistic Regression or XGBoost resulted in a noticeable degradation across all evaluation metrics compared to the baseline TabPFN model using the full feature set.

Conversely, applying TabPFN with the feature subset identified as most important by LightGBM yielded a marginal increase in accuracy (from 73.3% to 74.3%). However, this improvement was accompanied by an identical AUC score (0.835) and slight decreases in performance for Log-Loss (0.487 to 0.489) and AP (0.875 to 0.867). Analysis of the confusion matrices indicated that using this LightGBM-derived feature subset resulted in a miniscule improvement (Figure 10).

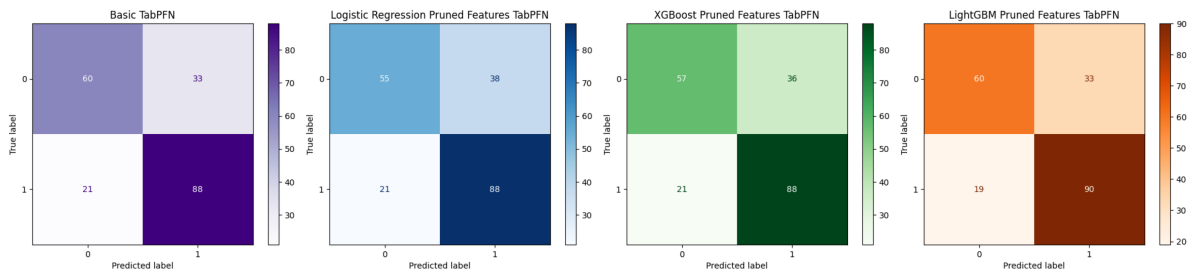


Figure 10: Comparison of confusion matrices on the test set for TabPFN using different feature subsets: (a) Basic TabPFN with all features, (b) TabPFN with Logistic Regression pruned features, (c) TabPFN with XGBoost pruned features, and (d) TabPFN with LightGBM pruned features. Note the slight reduction in false positives (top-right cell) in (d) compared to (a), as mentioned in the text. Values indicate counts for True Negatives, False Positives, False Negatives, and True Positives.

Model	Accuracy	AUC	Log-Loss	AP
Basic TabPFN	0.733	0.835	0.487	0.875
TabPFN with Logistic Regression pruned features	0.708	0.826	0.505	0.860
TabPFN with LightGBM pruned features	0.743	0.835	0.489	0.867
TabPFN with XGBoost pruned features	0.718	0.824	0.500	0.862

Table 4: Performance comparison of a basic and a feature subset trained models on the test set, evaluated using Accuracy, AUC, Log-Loss, and AP.

7 Conclusion

This project aimed to enhance the predictive accuracy of Counter-Strike 2 match outcomes by developing a novel feature representation strategy centred on short-term (90-day), map-specific player statistics and evaluating advanced machine learning models, including the Transformer-based TabPFN. Our methodology involved aggregating detailed player performance metrics into team-level summary statistics (mean, standard deviation, sum, minimum, maximum) and comparing the efficacy of tuned Logistic Regression, LightGBM, XGBoost, and TabPFN models.

Our findings strongly support the hypothesis that temporally granular, map-aware features significantly contribute to predictive power. The feature engineering approach successfully captured relevant signals, with metrics like kill/death ratio, team map win rates, deaths per round, and HLTV Rating 2.0 consistently identified as important predictors. The comparative model evaluation revealed that while tuned models improved upon baselines, feature selection via permutation importance was crucial for maximizing the performance of gradient boosting methods. Specifically, LightGBM trained on a pruned feature set achieved the highest overall performance, yielding a test accuracy of 79.7% and an AUC of 0.862, closely followed by the pruned XGBoost model.

The TabPFN model also demonstrated considerable promise, delivering competitive performance, particularly the best Log-Loss score, without requiring hyperparameter tuning or explicit feature selection, suggesting its robustness in handling the provided feature set. While using pruned features offered minimal benefit to TabPFN, its strong baseline performance validates its potential for esports analytics tasks. The performance levels achieved by our top models, especially pruned LightGBM, notably exceed the 60-65% accuracy benchmarks commonly reported in the cited literature for CS:GO prediction.

However, these promising results must be interpreted with caution due to a primary limitation: the relatively small dataset size (1010 matches) and restricted time frame (November 2023 - January 2024). The high observed accuracy might be specific to the competitive meta and team dynamics prevalent during this period and requires validation on larger, more temporally diverse datasets to ensure generalizability. Furthermore, the practical comparison against historical betting odds, initially planned as a benchmark, could not be fully realized due to data acquisition challenges.

Future work should prioritize validating these models and feature engineering strategies on expanded datasets covering longer time periods and potentially different tiers of play. Investigating additional feature types, such as in-game economic indicators or player role specializations as well as implementation of an Elo type system such as the TrueSkill 2 could offer further performance improvements. Finally, acquiring historical betting odds would enable a more direct assessment of the practical utility of these models for applications like algorithmic betting.

In summary, this study successfully demonstrated that combining short-term, map-specific feature aggregation with systematic model evaluation and feature selection can yield high-performance predictive models for CS2 match outcomes. While further validation is warranted, the results highlight valuable directions for esports analytics and showcase the potential of modern machine learning techniques applied to esports data.

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